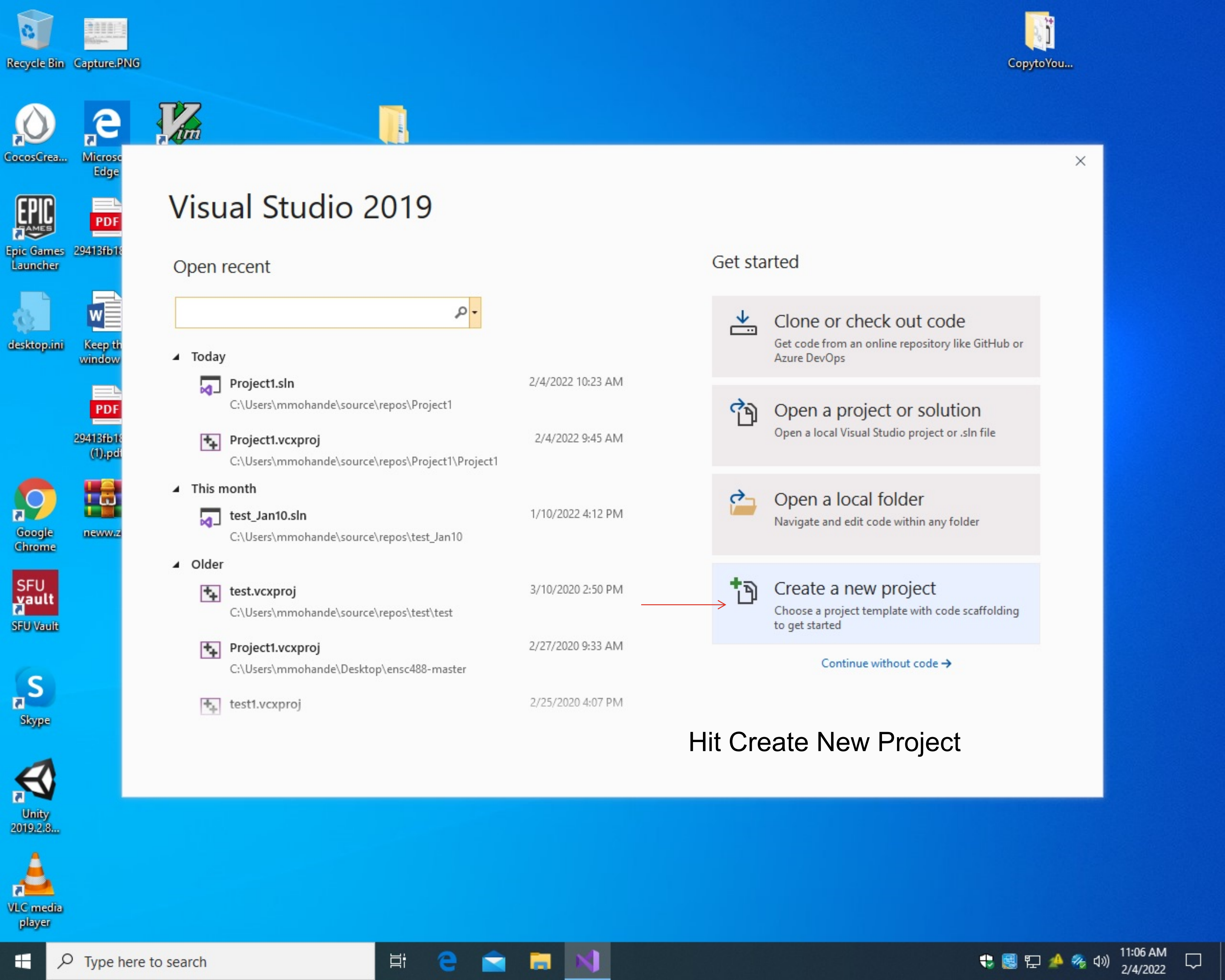


To setup the ENSC-488 emulator please follow the instructions in the following pages.

- The emulator is compatible with Visual Studio 2019 and 2017. (may be compatible with other versions but we have not tested)



Visual Studio 2019

Open recent

Today

-  **Project1.sln** 2/4/2022 10:23 AM
C:\Users\mmohande\source\repos\Project1
-  **Project1.vcxproj** 2/4/2022 9:45 AM
C:\Users\mmohande\source\repos\Project1\Project1

This month

-  **test_Jan10.sln** 1/10/2022 4:12 PM
C:\Users\mmohande\source\repos\test_Jan10

Older

-  **test.vcxproj** 3/10/2020 2:50 PM
C:\Users\mmohande\source\repos\test\test
-  **Project1.vcxproj** 2/27/2020 9:33 AM
C:\Users\mmohande\Desktop\ensc488-master
-  **test1.vcxproj** 2/25/2020 4:07 PM

Get started

 **Clone or check out code**
Get code from an online repository like GitHub or Azure DevOps

 **Open a project or solution**
Open a local Visual Studio project or .sln file

 **Open a local folder**
Navigate and edit code within any folder

 **Create a new project**
Choose a project template with code scaffolding to get started

[Continue without code →](#)

Hit Create New Project

Create a new project

Recent project templates

Empty Project

C++

Hit Empty Project
&
Next

Search for templates (Alt+S)



Clear all

C++

All platforms

All project types



Empty Project

Start from scratch with C++ for Windows. Provides no starting files.

C++

Windows

Console



Console App

Run code in a Windows terminal. Prints "Hello World" by default.

C++

Windows

Console



Windows Desktop Wizard

Create your own Windows app using a wizard.

C++

Windows

Desktop

Console

Library



Windows Desktop Application

A project for an application with a graphical user interface that runs on Windows.

C++

Windows

Desktop



Shared Items Project

A Shared Items project is used for sharing files between multiple projects.

C++

Windows

Android

iOS

Linux

Desktop

Console

Library

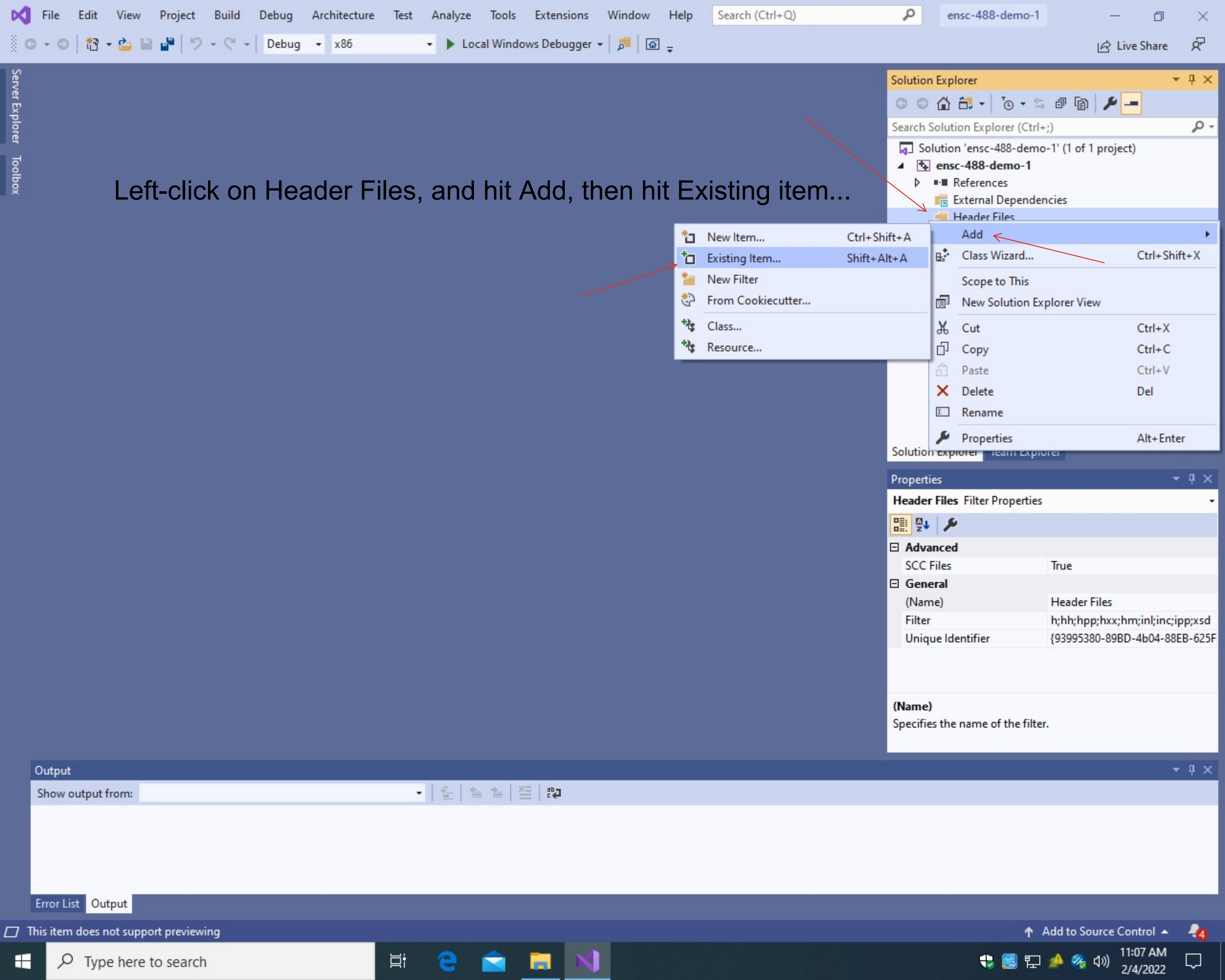
UWP

Games

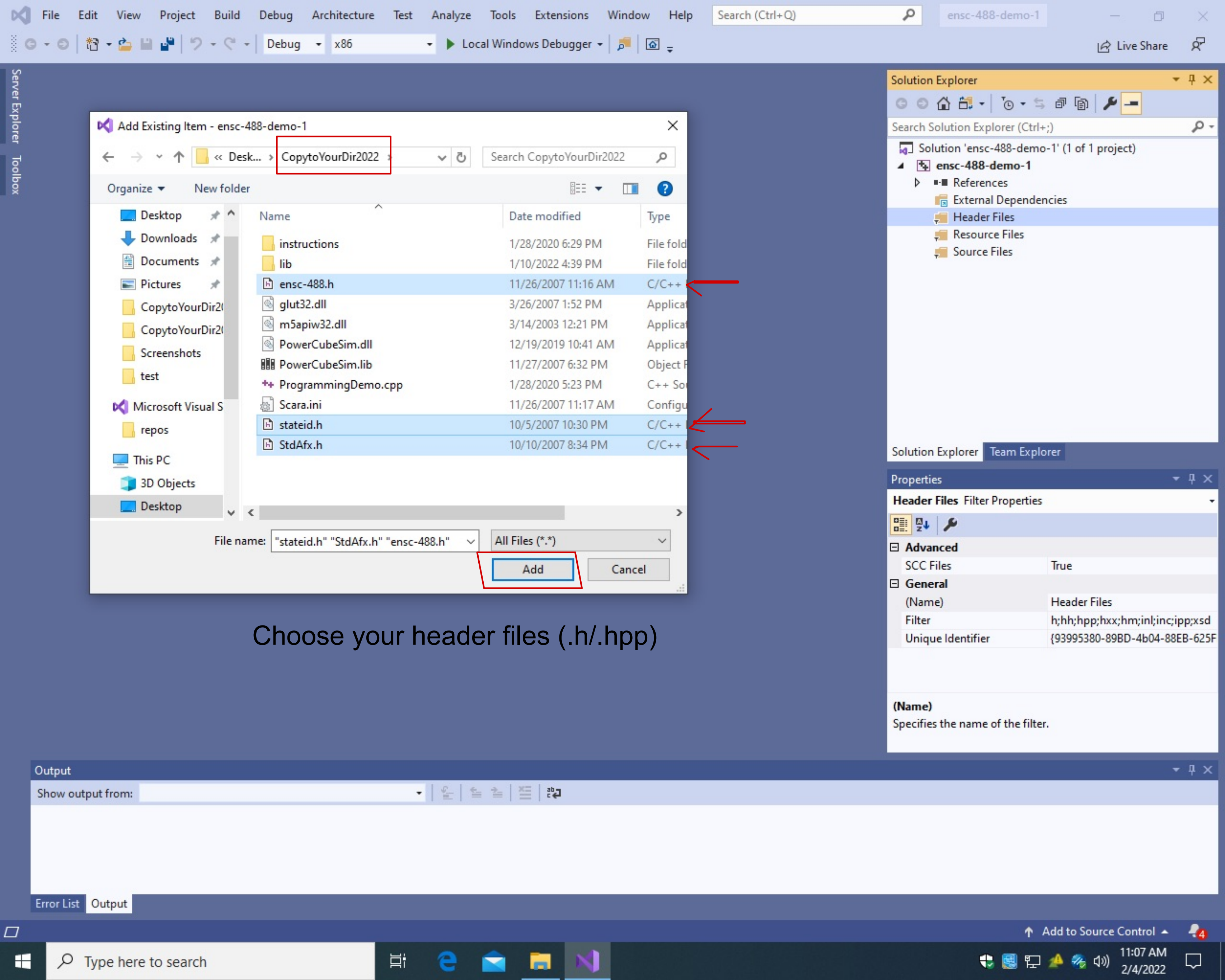
Mobile

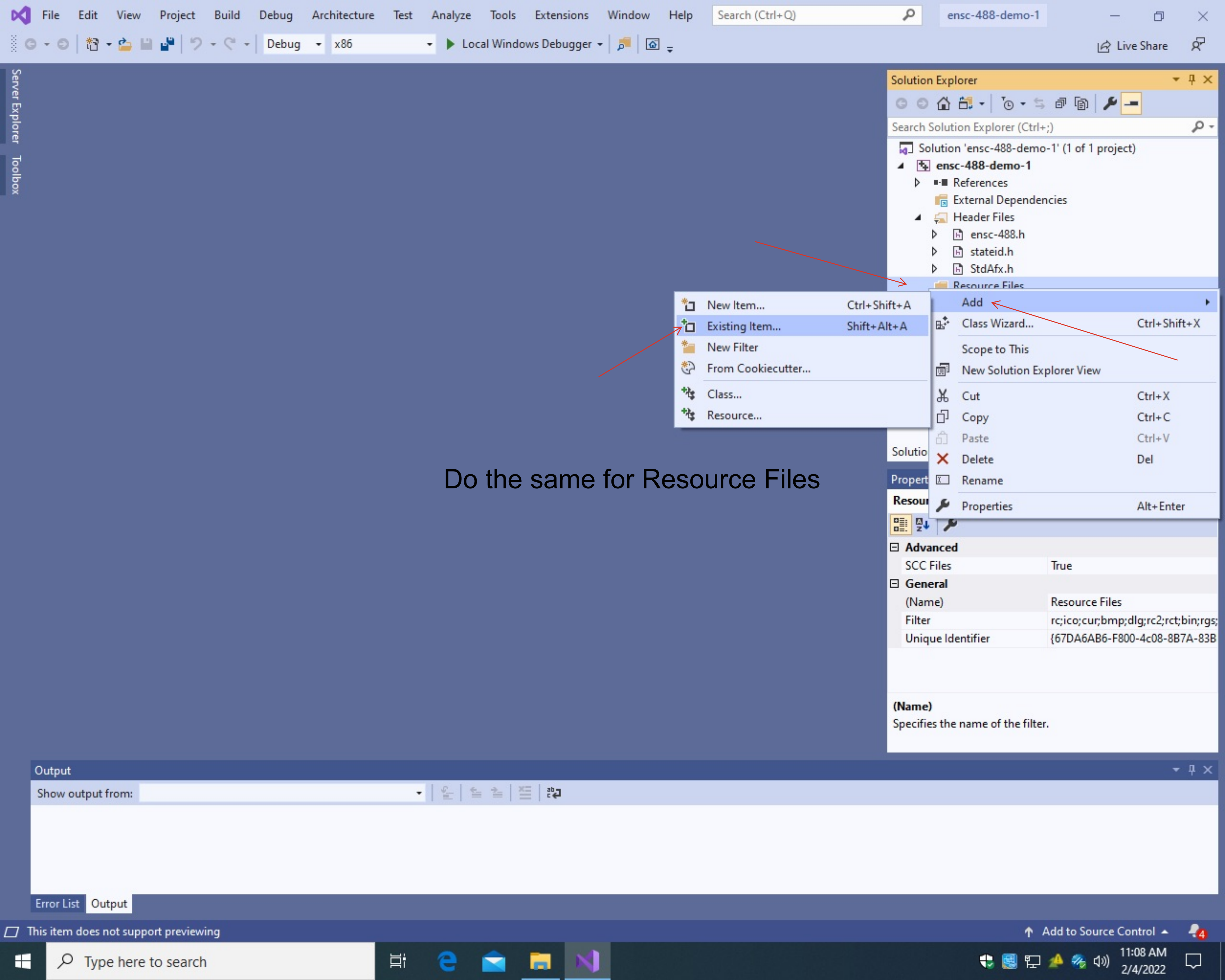
Back

Next

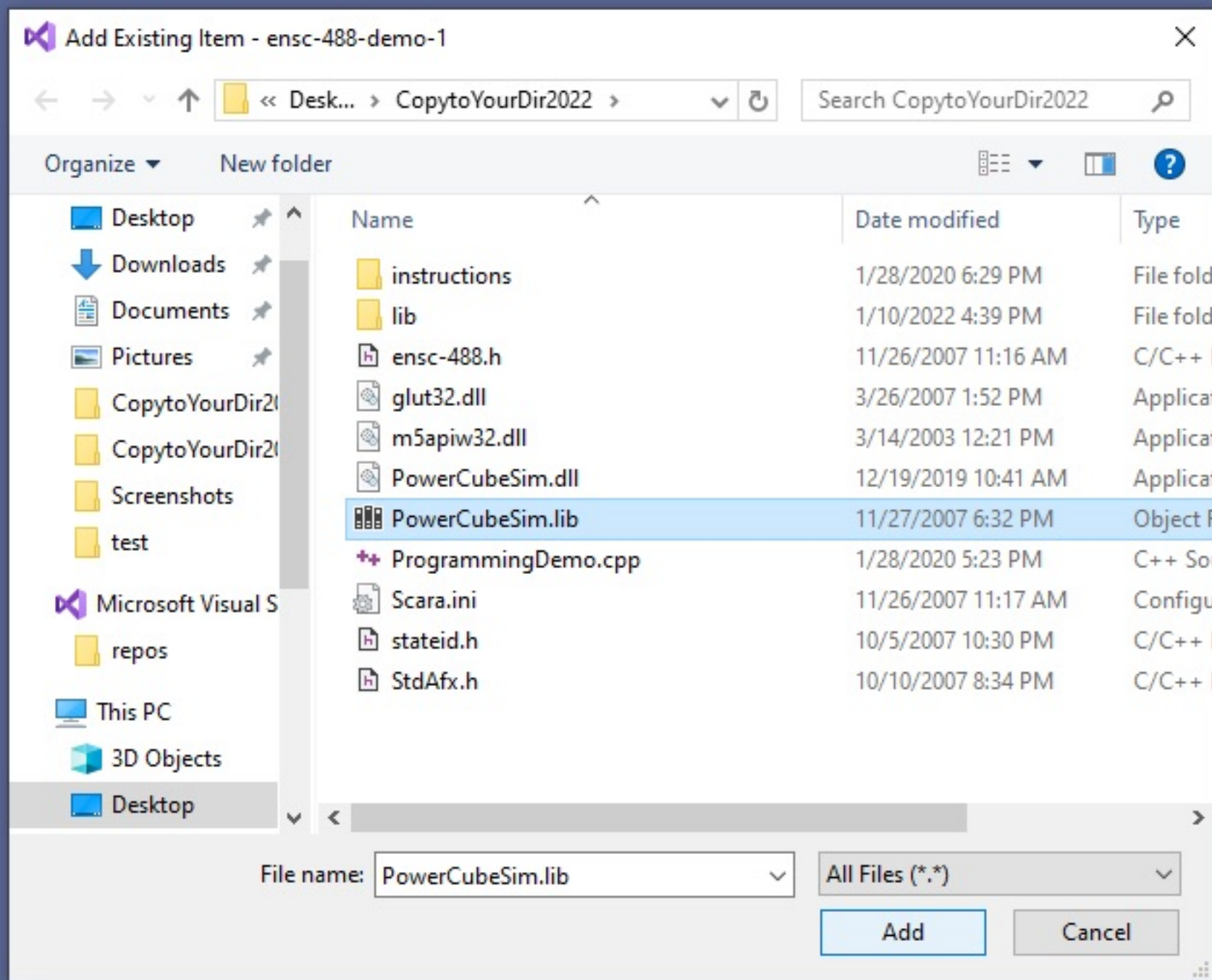


Left-click on Header Files, and hit Add, then hit Existing item...

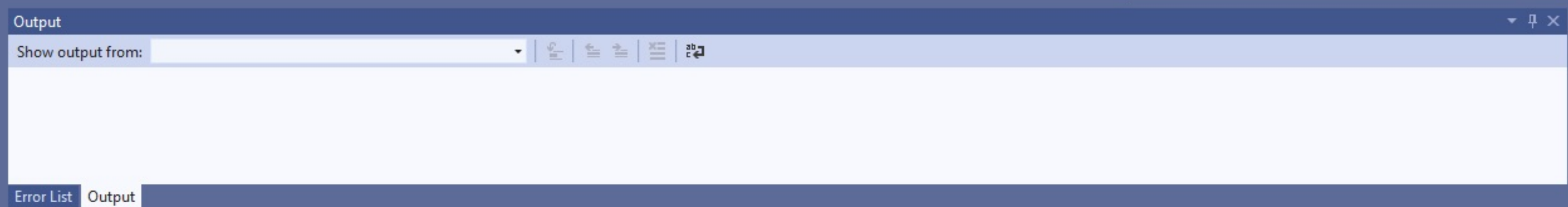
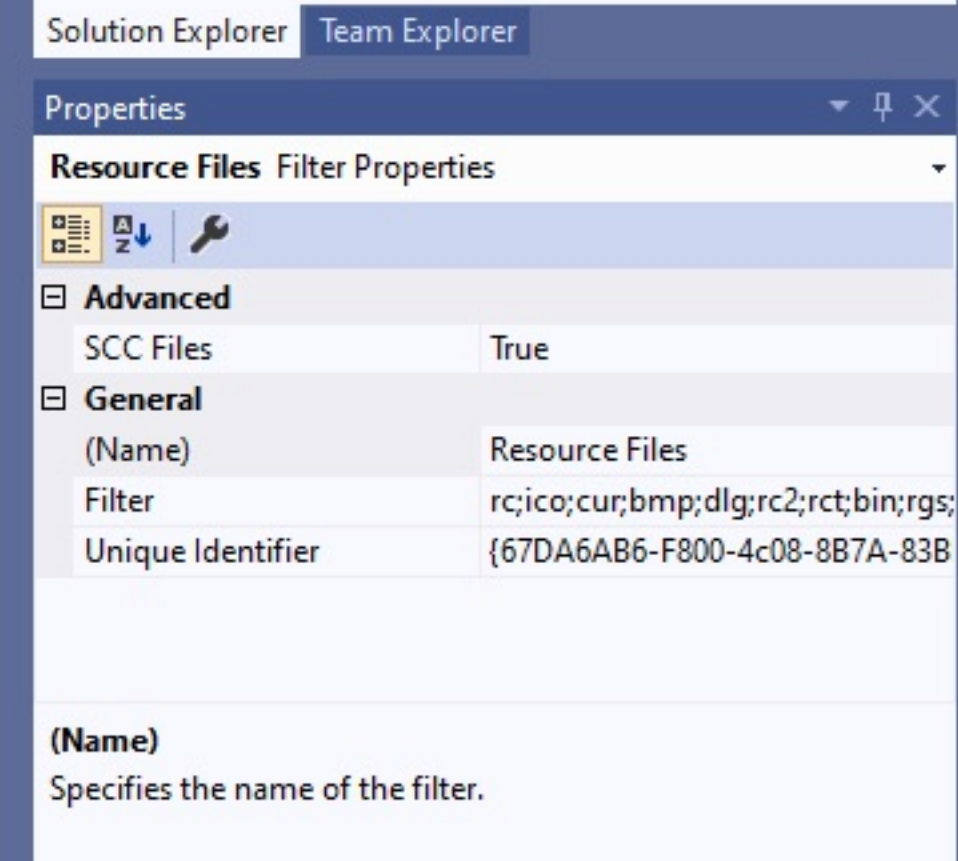
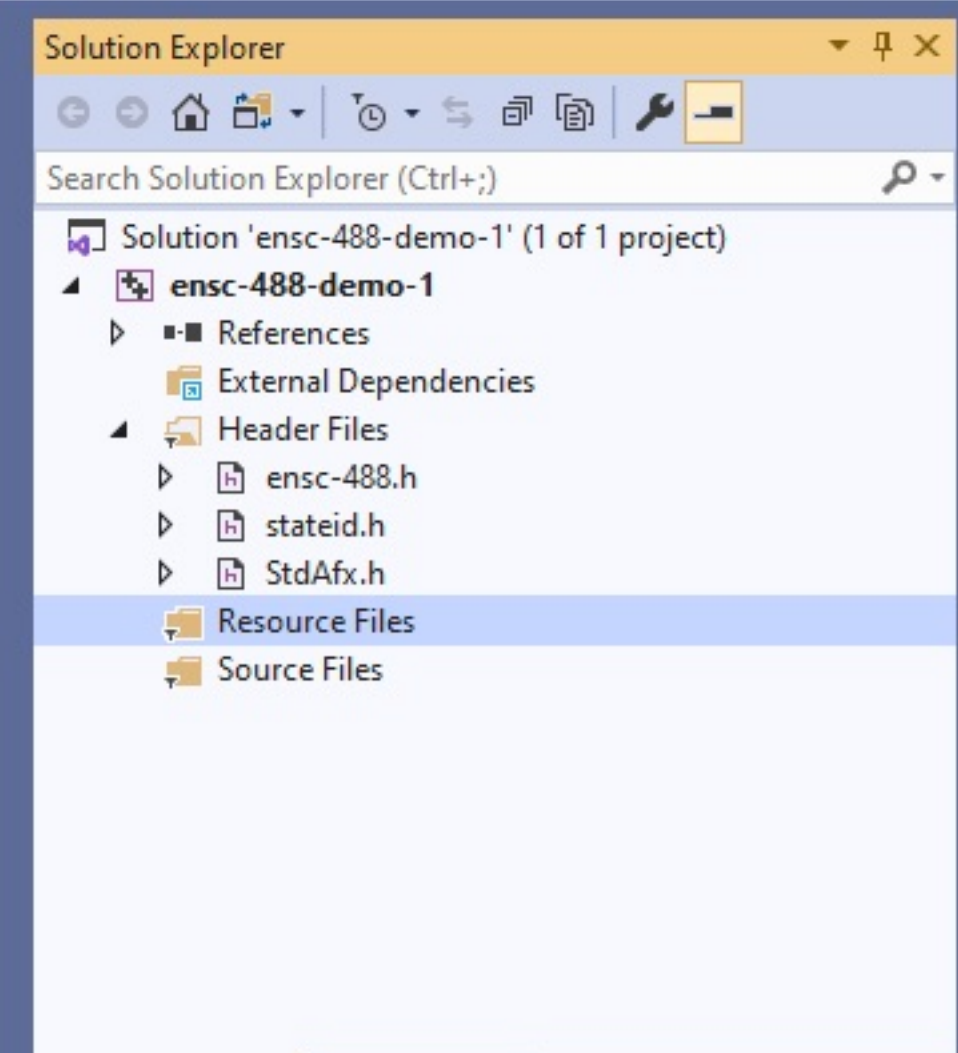


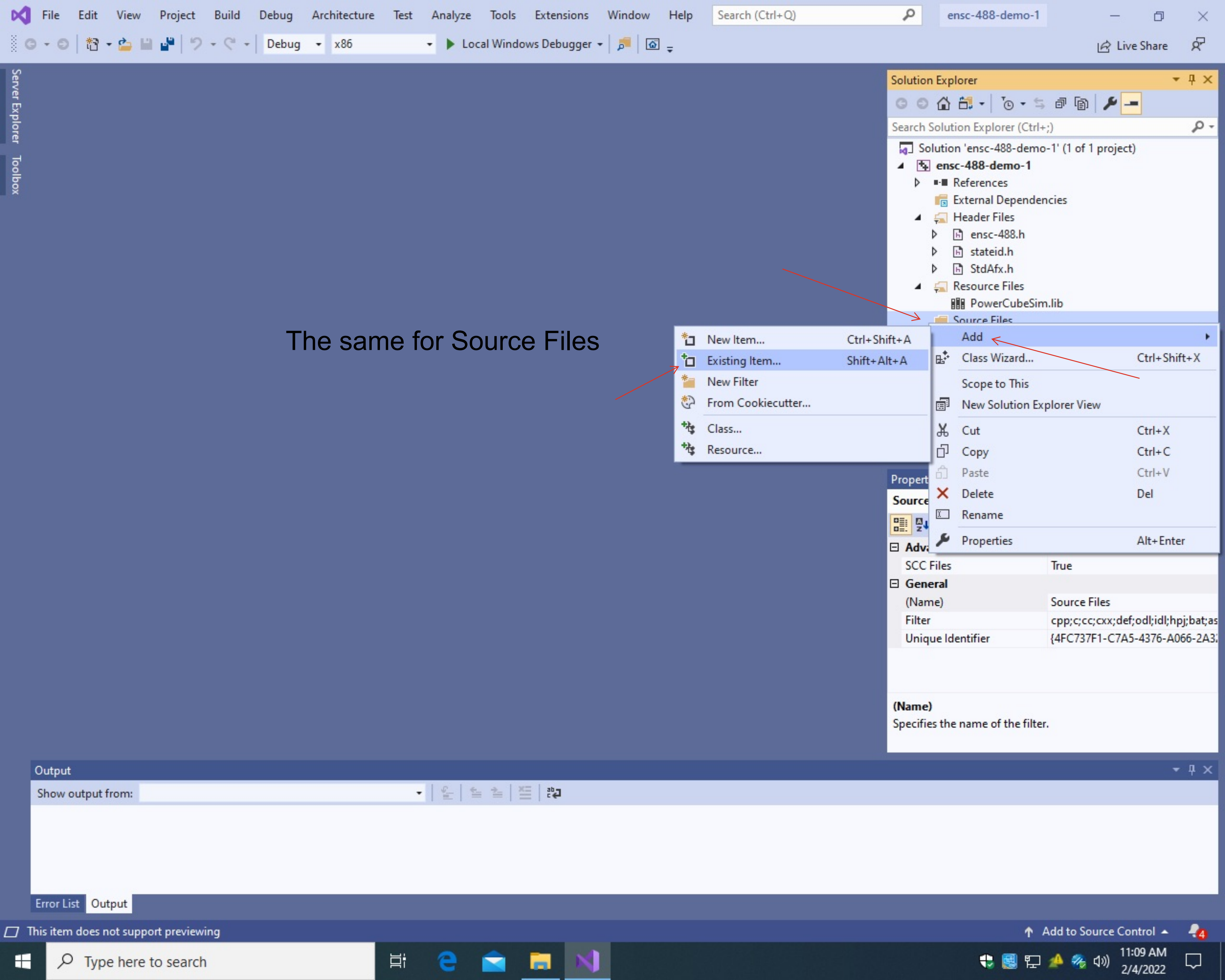


Do the same for Resource Files



Select PowerCubeSim.lib





The same for Source Files

Context menu for Source Files:

- New Item... (Ctrl+Shift+A)
- Existing Item... (Shift+Alt+A)
- New Filter
- From Cookiecutter...
- Class...
- Resource...

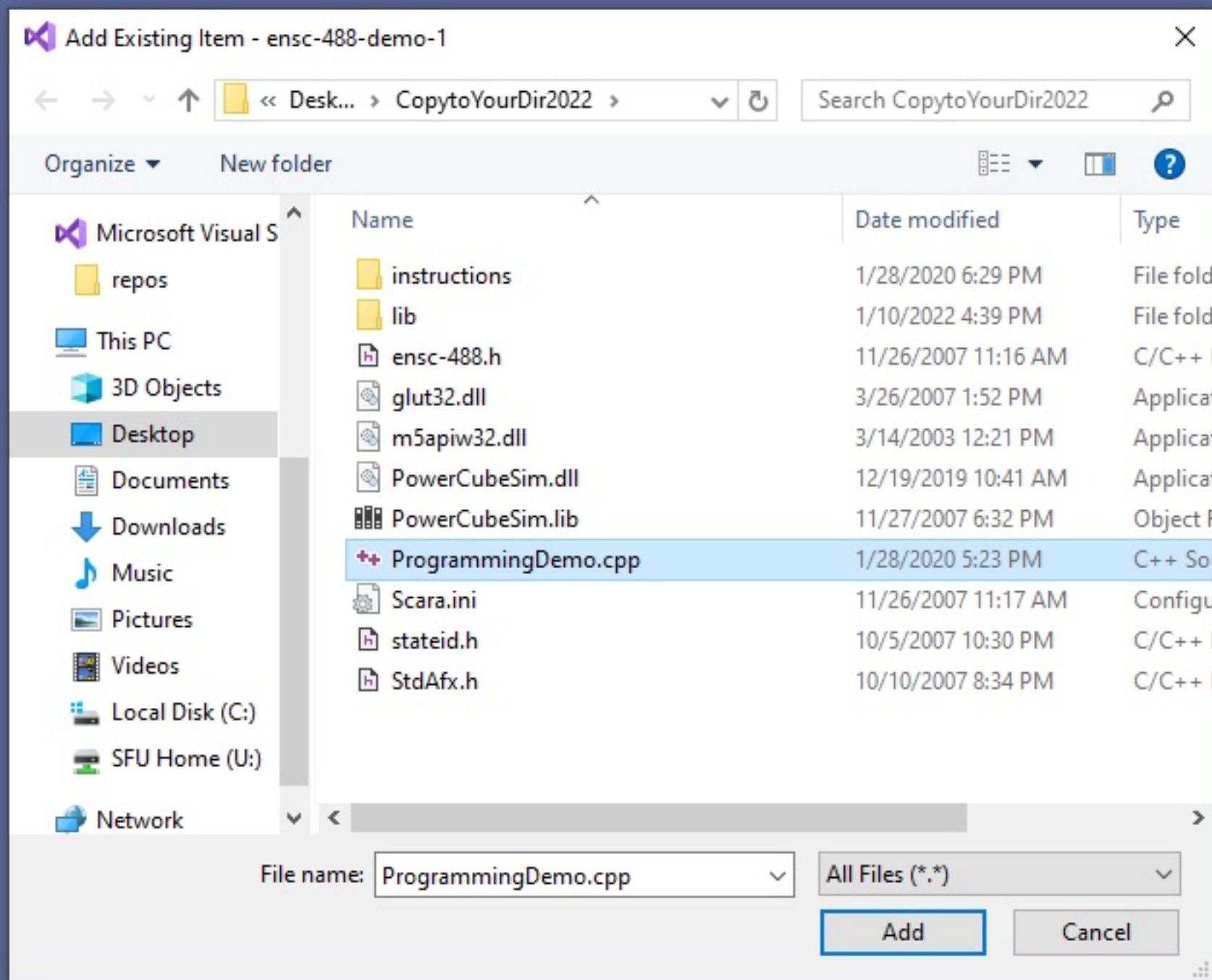
Submenu for Add:

- Add
- Class Wizard... (Ctrl+Shift+X)
- Scope to This
- New Solution Explorer View
- Cut (Ctrl+X)
- Copy (Ctrl+C)
- Paste (Ctrl+V)
- Delete (Del)
- Rename
- Properties (Alt+Enter)

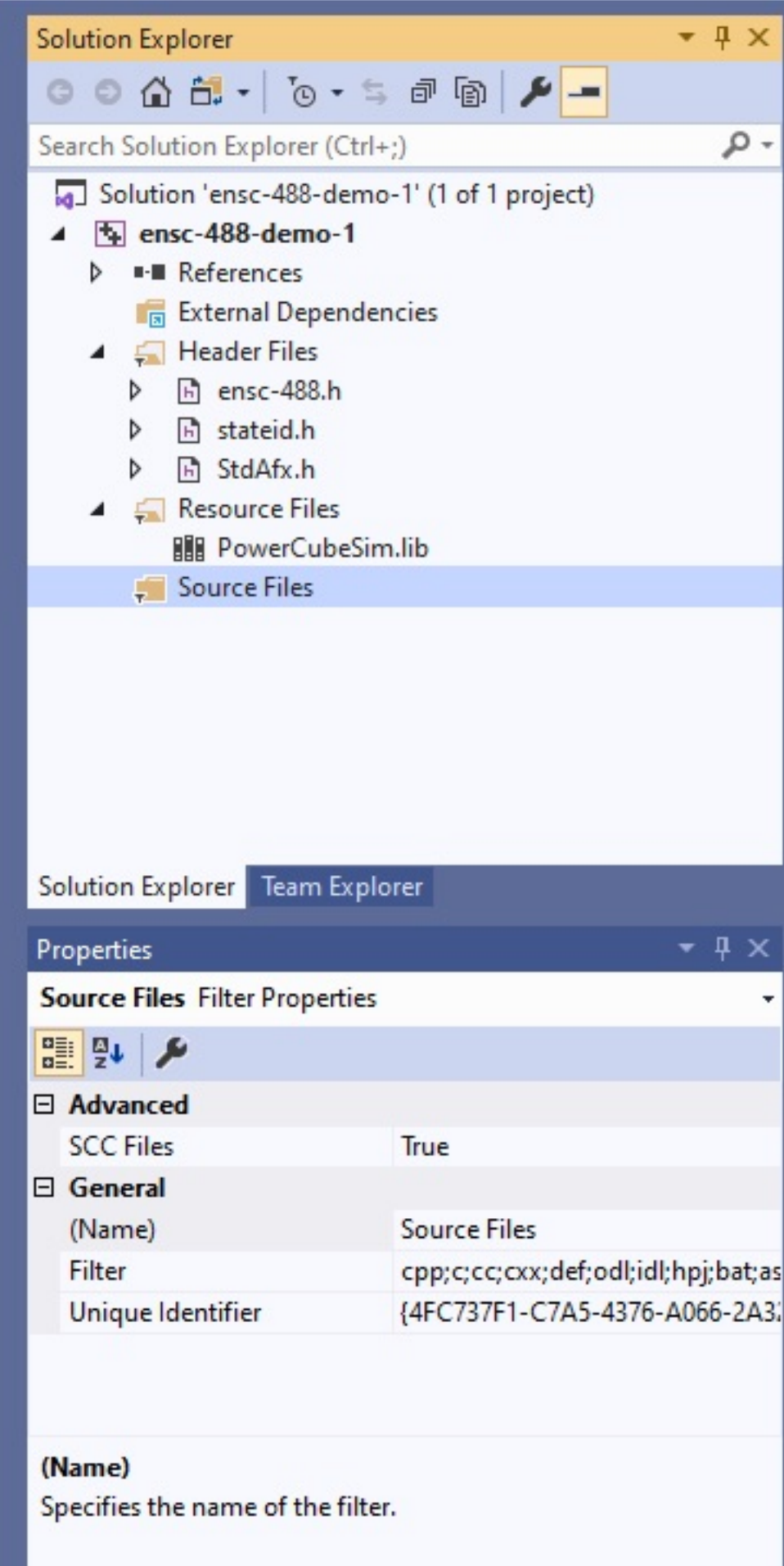
Properties window - General tab:

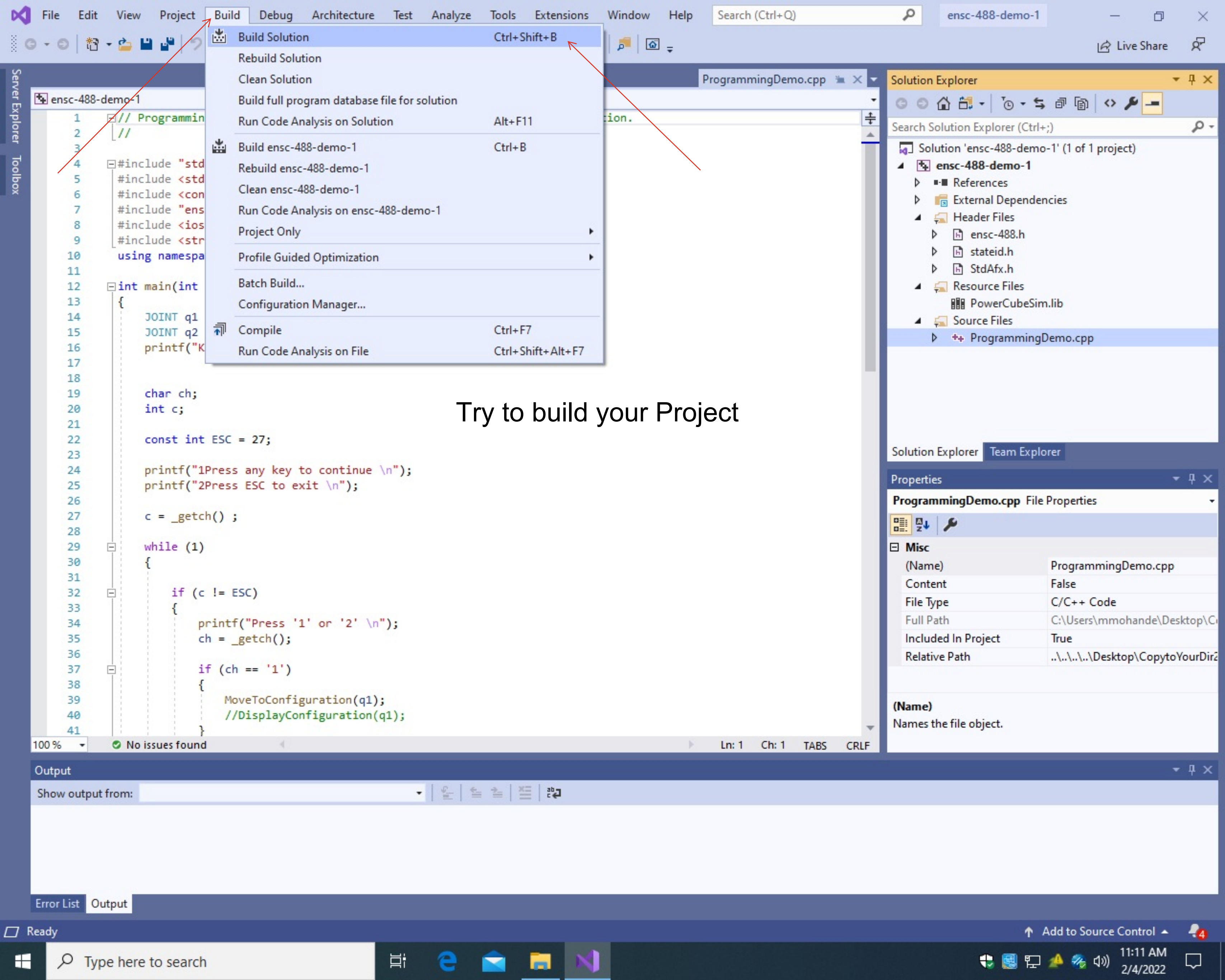
SCC Files	True
(Name)	Source Files
Filter	cpp;c;cc;cxx;def;odl;idl;hpj;bat;as
Unique Identifier	{4FC737F1-C7A5-4376-A066-2A3}

(Name)
Specifies the name of the filter.

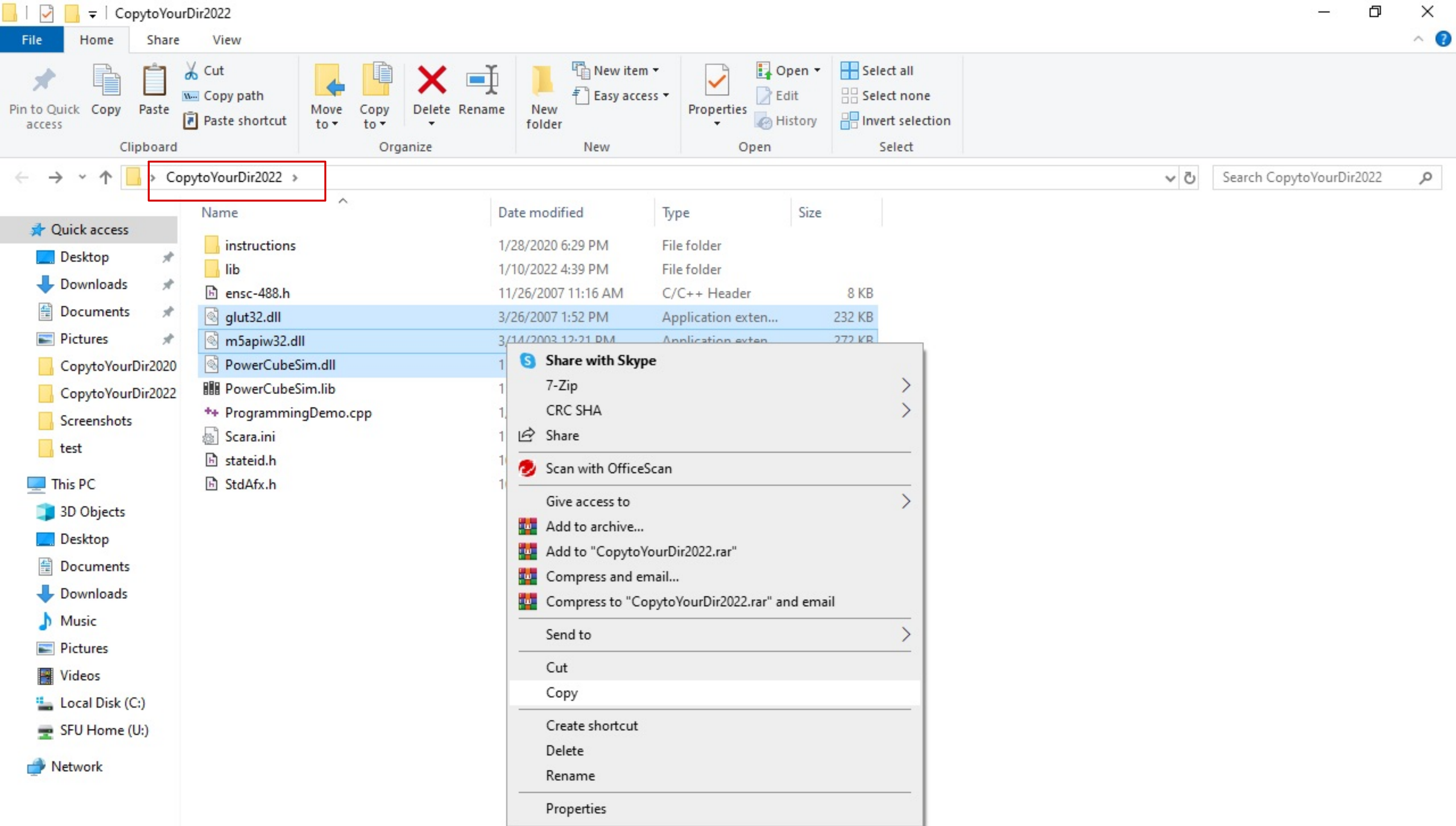


Choose your source files (.c/.cpp)

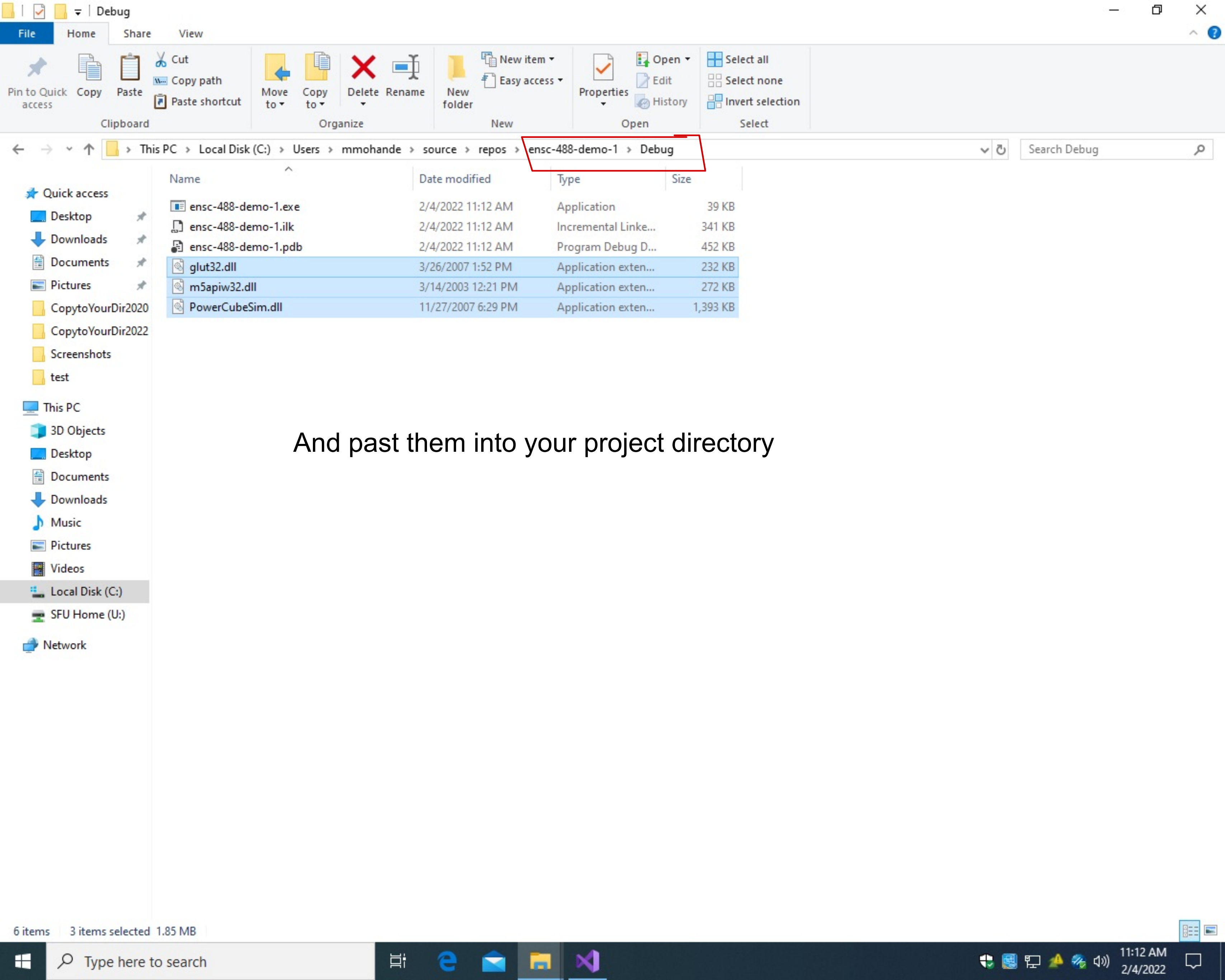




Try to build your Project



Copy these 3 .dll files from the extracted .zip file



For Visual Studio 2022 (or above)

The Visual Studio Project must be setup to compile in **Win32 mode**.
By default, VS2022 selects x64 as the target platform, and this will cause the compiler to complain.

If you intend to use VS > 2019 & if VS complained follow this page

If you receive the following error in Visual Studio 2022 (or above):

 LNK2019 unresolved external symbol "__declspec(dllimport) bool __cdecl MoveToConfiguration(double (&)[4],bool)" (__imp_?MoveToConfiguration@@YA_NAEAY03N_N@Z) referenced in function main	Project1	ProgrammingDemo.obj	1
 LNK4272 library machine type 'x86' conflicts with target machine type 'x64'	Project1	PowerCubeSim.lib	1
 LNK1120 1 unresolved externals	Project1	Project1.exe	1

Configure the project's target to compile using Win32 as the platform (for VS2022+ versions only):

- 1) Go To: Project->properties->configuration manager.
- 2) Change the project target to Win32

FileEditViewGitProjectBuildDebugTestAnalyzeToolsExtensionsWindowHelpSearch (Ctrl+Q)

ensc488-demo-1Sign in

Live Share

Debugx64Local Windows Debugger

PowerCubeSim.lib Property Pages

Configuration: All ConfigurationsPlatform: Win32Configuration Manager...

Configuration Properties

General

General

Excluded From Build

Content

Item Type

Output

Show output from: Build

Build started...
1>----- Build started: Project: e
1>ProgrammingDemo.cpp
1>ensc488-demo-1.vcxproj -> C:\Use
===== Build: 1 succeeded, 0 f
===== Elapsed 00:02.615 =====

Configuration Manager

Active solution configuration: DebugActive solution platform: x64

Project contexts (check the project configurations to build or deploy):

Project	Configuration	Platform	Build	Deploy
ensc488-demo-1	Debug	Win32	☒	☐

Error ListOutput

Build succeeded

Close

FileEditViewProjectBuildDebugArchitectureTestAnalyzeToolsExtensionsWindowHelp

Search (Ctrl+Q)

ensc-488-demo-1

Live Share

Debugx86Local Windows DebuggerLocal Windows Debugger

ProgrammingDemo.cpp

ensc-488-demo-1(Global Scope)

```
1 // ProgrammingDemo.cpp : Defines the entry point for the console application.
2 //
3
4 #include "stdafx.h"
5 #include <stdio.h>
6 #include <conio.h>
7 #include "ensc-488.h"
8 #include <iostream>
9 #include <string>
10 using namespace std;
11
12 int main(int argc, char* argv[])
13 {
14     JOINT q1 = {0, 0, -100, 0};
15     JOINT q2 = {90, 90, -200, 45};
16     printf("Keep this window in focus, and...\n");
17
18
19     char ch;
20     int c;
21
22     const int ESC = 27;
23
24     printf("1Press any key to continue \n");
25     printf("2Press ESC to exit \n");
26
27     c = _getch() ;
28
29     while (1)
30     {
31
32         if (c != ESC)
33         {
34             printf("Press '1' or '2' \n");
35             ch = _getch();
36
37             if (ch == '1')
38             {
39                 MoveToConfiguration(q1);
40                 //DisplayConfiguration(q1);
41             }
42         }
43     }
44 }
```

Finally Run the Project!

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'ensc-488-demo-1' (1 of 1 project)

ensc-488-demo-1

References

External Dependencies

Header Files

ensc-488.h

stateid.h

StdAfx.h

Resource Files

PowerCubeSim.lib

Source Files

ProgrammingDemo.cpp

Solution ExplorerTeam Explorer

Properties

ProgrammingDemo.cpp File Properties

Misc

(Name)	ProgrammingDemo.cpp
Content	False
File Type	C/C++ Code
Full Path	C:\Users\mmohande\Desktop\C...
Included In Project	True
Relative Path	..\..\..\Desktop\CopytoYourDir2

(Name)

Names the file object.

Output

Show output from: Build

1>----- Build started: Project: ensc-488-demo-1, Configuration: Debug Win32 -----
1>ProgrammingDemo.cpp
1>ensc-488-demo-1.vcxproj -> C:\Users\mmohande\source\repos\ensc-488-demo-1\Debug\ensc-488-demo-1.exe
===== Build: 1 succeeded, 0 failed, 0 up-to-date, 0 skipped =====

Error ListOutput

Build succeeded

Add to Source Control

11:13 AM 2/4/2022

